

Scholium

Structural Analogy with Ricci Flow and Perelman's Surgery

English Translation and Standalone Version

V.F.S. v2.0 / Open-Gate

Supplement to V.F.S. – Differential Geometry of Filtrum Lucis

Scholium

The mathematical architecture of V.F.S. displays a deep structural affinity with the differential-geometric apparatus associated with Grigori Perelman and his proof of the Poincare conjecture through Ricci flow with surgery. The analogy covers a set of structural correspondences with different levels of strictness. Before listing them systematically, one must state the main limitation clearly: **V.F.S. is not an imitation of Perelman's theory**. It has its own formal architecture, including features absent from the original Ricci-flow framework. The analogy is inspiring, not derivational.

This Scholium compares **structural roles**, not mathematical objects. Ricci flow evolves Riemannian metrics on manifolds; V.F.S. evolves symbolic-theological state variables. The tensor R_{ij} is a curvature tensor, whereas σ is a scalar resistance variable. Perelman's functionals $\mathcal{F}$ and $\mathcal{W}$ are geometric entropy functionals, whereas λ is Sophia, an internal wisdom variable of the model. The analogy is therefore **formal-functional**, not ontological and not theorem-equivalent.

1 Basic Construction of the Affinity

The closed conservation and the open balance of V.F.S. are:

$$\sigma + \lambda = C_0 \quad (\zeta_0 = 0), \quad (1)$$

$$\sigma + \lambda = C_0 + \mathcal{G}_{\text{recepta}}(t), \quad \dot{\sigma} + \dot{\lambda} \geq 0 \quad (\zeta_0 > 0). \quad (2)$$

In the closed limit ($\zeta_0 = 0$), this is an exact invariant. In the full Open-Gate model, it becomes an open balance law:

$$\frac{d}{dt}(\sigma + \lambda) = I_{\text{gate}}(t) \geq 0. \quad (3)$$

Thus it is not an invariant in the open case. Ricci flow has no exact analogue of such a two-variable balance; Perelman's $\mathcal{F}$ and $\mathcal{W}$ provide control through monotonicity rather than through conservation.

This is the first proper feature of V.F.S.: ontological mass, represented by the positive Imago Dei foundation $\mathcal{E}_0 > 0$, ensures that the core is not annihilated. In the closed limit, the pair $\sigma \leftrightarrow \lambda$ forms a conservative symmetry of transmutation. In the open model, the same structure

becomes an open balance: resistance is transmuted into Sophia, and Sophia also receives grace through the gate.

Within the physically meaningful domain of the model, no spiritual reality is annihilated; it is transformed. Evil has no independent being: it is a topological deformation which, through grace and human participation, can be transmuted into a more perfect form, in accordance with the logic of *privatio boni*.

The sharpened Sophia law is:

$$\dot{\lambda} = (\delta u - \gamma) \tanh(\kappa\sigma) + I_{\text{gate}}(t), \quad I_{\text{gate}}(t) \geq 0. \quad (4)$$

In the closed Microcosm limit ($I_{\text{gate}} = 0$), λ is monotone only conditionally, depending on whether u is above or below Λ_c . In the Open-Gate model, the bounded grace-source gives an open balance rather than unconditional monotonicity. When $u < \Lambda_c$,

$$\dot{\lambda} \geq 0 \iff I_{\text{gate}}(t) \geq (\gamma - \delta u) \tanh(\kappa\sigma). \quad (5)$$

Perelman's entropy imposes geometric irreversibility; λ preserves freedom. Sophia may grow, pause, or regress unless the grace-source dominates.

2 Strong Analogies: Shared Form and Shared Function

2.1 Geometric Evolution

Ricci flow describes the evolution of a metric,

$$\frac{\partial g_{ij}}{\partial t} = -2R_{ij}, \quad (6)$$

while V.F.S. describes the evolution of Being,

$$S = \frac{dP}{dt}. \quad (7)$$

Both are first-order evolutionary architectures in time. Both describe a geometry moving toward a canonical form: in Ricci flow, toward geometric decomposition and canonical model pieces; in V.F.S., toward Pleroma Christi.

2.2 Critical Localization Near Transition

Near singular points, both flows reveal localization of critical structure, a universal feature of transitions. In Ricci flow, high-curvature regions localize near singularity formation and are studied, after rescaling, through canonical high-curvature models. In V.F.S., near the Cross $x = 0$, the curvature of Filtrum Lucis reaches its maximum:

$$\Phi''(0) = \frac{k}{4}. \quad (8)$$

Thus the Cross $x = 0$ is the point of maximal transmissive curvature of Filtrum Lucis. The analogy is not a literal slowdown-by-equation, but the localization of critical structure. This corresponds conceptually to the Metastable Cathartic Layer: external stagnation may hide the inner accumulation of Sophia while the system hesitates near the transition point.

2.3 Irreversibility

In Ricci flow, after surgery, the topology or geometric continuation of the manifold has changed: an exact return to the previous geometry is not possible. In V.F.S., after the transmutation $\sigma \rightarrow \lambda$ in the regime of Transformation, the trajectory remembers the operation and cannot simply return to the old geometry. This is the Hysteresis of Salvation,

$$H = \int \sigma(u) du. \quad (9)$$

In the regime of Collapse, partial regression $\lambda \rightarrow \sigma$ may occur, but the hysteresis integral itself remains cumulative: it records the trace of transformations, not merely the instantaneous state.

2.4 Singularity as Transition, Not End

The deepest common feature is that singularity is not terminal. In Ricci flow, singularity initiates surgery and the flow continues. In V.F.S., death-like closure initiates Anastasis and participation continues. Both frameworks reject the finality of singular events.

3 Functional Analogies: Different Form, Shared Role

3.1 Surgery: Discrete vs Smooth

Perelman's surgery is a discrete procedure. When curvature approaches a singularity, the flow is stopped, the pathological region, often a neck pinch, is cut away, and the manifold is reassembled. In V.F.S., the surgery of σ is smooth and endogenous. It is built into the equations through the factor $\tanh(\kappa\sigma)$, without stopping the flow or introducing an external cut.

The formal mechanisms differ, but the function is similar: transformation instead of collapse. The term $\delta\sqrt{VF} + \varepsilon$ may be read as the mathematical image of the conjunction between human will, human action, and the action of grace that smooths the singular resistance σ .

3.2 Sophia vs Perelman's Entropy

Perelman's entropy functionals are monotone within the appropriate Ricci-flow and conjugate heat-equation framework; this gives one-way control of the geometric flow. Sophia satisfies:

$$\frac{d\lambda}{dt} = (\delta\sqrt{VF} + \varepsilon - \gamma) \tanh(\kappa\sigma) + I_{\text{gate}}(t), \quad (10)$$

where

$$I_{\text{gate}}(t) = \zeta_0 g_{\text{eff}}(t) e^{-\phi\sigma} (1 + \chi\lambda_+)^{-1} \geq 0. \quad (11)$$

In the closed limit, monotonicity is conditional and requires $\sqrt{VF} + \varepsilon \geq \Lambda_c$. In the open model, Sophia has a second non-negative source, so the total derivative may remain positive even when the transmutation term is weak.

This is not a defect of the model, but a second proper feature of V.F.S.: it leaves room for freedom. Unconditional monotonicity would make the human subject an automaton of theosis. Conditional monotonicity with bounded grace-source preserves freedom while still allowing grace to intervene.

3.3 Epektasis vs Ricci Normalization

This is the deepest functional analogy, because it has the opposite formal direction. Normalized Ricci flow adds a scale-correcting term in order to preserve volume and prevent degeneration by collapse of scale. V.F.S. does the opposite through Epektasis:

$$\dot{\Omega}_P = \alpha\lambda. \quad (12)$$

In the regime of sustained Sophia, the Brim of the Vessel expands rather than remaining fixed. The exact statement is:

$$\Omega_P(t) = \Omega_P(0) + \alpha \int_0^t \lambda(s) ds. \quad (13)$$

Thus

$$\Omega_P(t) \rightarrow \infty \iff \int_0^\infty \lambda(t) dt = +\infty. \quad (14)$$

If additionally $\lambda(t) \rightarrow \lambda_\infty > 0$, then

$$\Omega_P(t) \sim \alpha\lambda_\infty t. \quad (15)$$

Both systems address the same problem - avoiding degenerative behavior and allowing the flow to continue - but by opposite means: one controls scale, the other opens scale.

3.4 $K(t)$ as Singularity Detector

In Ricci flow, singularity is detected through blow-up of curvature derivatives, for example $|\nabla^k R| \rightarrow \infty$. In V.F.S., catharsis is detected through the sign-change of a finite function:

$$\Phi'''(0) = 0. \quad (16)$$

These are very different mathematical mechanisms - unbounded growth versus controlled sign-change - but they serve an analogous role: they identify a critical point. The V.F.S. mechanism is softer mathematically and more explicitly theological: the Cross is not an explosion, but the point where the sign of being is inverted.

4 Heuristic Analogies

4.1 σ as Curvature of Being

Ricci curvature R_{ij} is a tensor and may blow up. The variable σ is a scalar resistance variable. In the closed limit,

$$\sigma + \lambda = C_0, \quad (17)$$

and in the Open-Gate model,

$$\sigma + \lambda = C_0 + \mathcal{G}_{\text{recepta}}(t), \quad \dot{\sigma} + \dot{\lambda} \geq 0. \quad (18)$$

Inside the active domain of V.F.S., σ is controlled and non-negative. The factor $\tanh(\kappa\sigma)$ protects the lower boundary $\sigma = 0$, while upper boundedness belongs to the active-domain and Lyapunov-corridor assumptions. Unlike Ricci-curvature blow-up, σ is not used as an unbounded singularity detector inside the active V.F.S. proof.

4.2 Anastasis as Post-Surgical Reassembly

In Ricci flow, after surgery, the manifold continues with a reassembled regular geometry. In V.F.S., Anastasis is associated with the second sign-change of the Filtrum Lucis derivative structure, namely the zero of $\Phi^{(4)}$ at the positive threshold $x = x_+$. The shared meaning is continuation after a critical event; the mathematical form is different.

5 Proper Features of V.F.S. Not Found in Ricci Flow

5.1 Paired Anastasis Structure

The fourth derivative $\Phi^{(4)}$ has two symmetric zeros around the zero of Φ''' , the Cross:

$$x_{\pm} = \pm \frac{1}{k} \ln(2 + \sqrt{3}), \quad x_- + x_+ = 0. \quad (19)$$

This is the Paschal Triad: Good Friday, the entrance into death; Holy Saturday, the Cross at the center; and Resurrection Sunday, the exit from death. The Cross is the geometric center of symmetry. Ricci flow has no analogue of this paired Paschal structure.

5.2 Closed Conservation and Open-Gate Balance

Perelman's framework has no law of the form "sum of two quantities equals a constant." In the closed limit ($\zeta_0 = 0$), V.F.S. does:

$$\sigma + \lambda = C_0. \quad (20)$$

In the full Open-Gate model, this becomes:

$$\sigma + \lambda = C_0 + \mathcal{G}_{\text{recepta}}(t), \quad \dot{\sigma} + \dot{\lambda} \geq 0. \quad (21)$$

This is an open balance rather than a Noether invariant in the strict sense, since no Lagrangian and no continuous symmetry of an action have been specified.

5.3 Free Sophia

The possibility of the regression of λ when $u < \Lambda_c$ is the mathematical trace of freedom. In the open model, regression is not governed by synergy alone: the bounded gate source $I_{\text{gate}} \geq 0$ may compensate for it or overcome it. Freedom remains real, but grace is not absent in weak synergy.

5.4 Filtrum Lucis as a Single Analytic Function

Filtrum Lucis is given by a single analytic softplus-type function:

$$\Phi(x) = \frac{1}{k} \ln(1 + e^{kx}). \quad (22)$$

Its derivatives $\Phi', \Phi'', \Phi''', \Phi^{(4)}, \dots$ are algebraic combinations of the sigmoid $\text{sig}(kx)$. The zero-structure of these derivatives encodes the theological hierarchy. Ricci flow has no analogue of such a single filtering function.

5.5 Manere ε and Imago Dei $\mathcal{E}_0$

The constants ε and $\mathcal{E}_0$ act as anti-collapse structures. They prevent algebraic disappearance of Pleroma wherever the formula for P is defined:

$$P(t) \geq \mathcal{E}_0. \quad (23)$$

This does not mean that active motion cannot collapse dynamically. It means that the Imago Dei foundation prevents the annihilation of the Pleroma core, while active participation remains dependent on the human $V \cdot F$ dynamics.

6 Summary Table of the Analogies

#	Ricci Flow / Perelman	V.F.S.	Level
I	$\partial_t g_{ij} = -2R_{ij}$; metric evolution	$S = dP/dt$; evolution of Being	strong
II	Curvature tensor R_{ij} ; possible blow-up	Scalar resistance σ ; active-domain controlled	heuristic
III	Discrete surgery / excision	Smooth surgery of σ / transmutation	functional
IV	Entropy $\mathcal{F}, \mathcal{W}$; monotone control	Sophia λ ; closed conditional monotonicity and open bounded-source balance	functional
V	Monotone control / regulatory structure	$\sigma + \lambda = C_0 + \mathcal{G}_{\text{recepta}}(t)$; open balance	heuristic
VI	Normalization preserves scale or volume	Epektasis expands capacity: $\dot{\Omega}_P = \alpha\lambda$	functional, opposite form
VII	Singularity detector: $ \nabla^k R \rightarrow \infty$	Catharsis $K(t)$; sign-change $\Phi'''(0) = 0$	functional
VIII	Post-surgical reassembly	Anastasis; second zero of $\Phi^{(4)}$	heuristic
IX	Critical localization near singularity	Metastable Cathartic Layer near $x = 0$	functional
X	Irreversibility after surgery	Hysteresis of Salvation $H = \int \sigma du$	functional / heuristic
XI	Singularity as transition, not end	Cross as transition, not end	strong
XII	No direct analogue	Open Gate g_{eff} : guaranteed openness becomes acquired receptivity	proper V.F.S. feature

The hysteresis analogy is strong existentially but heuristic mathematically: Ricci surgery changes topology or geometric continuation, whereas V.F.S. hysteresis is a model-defined functional of memory.

7 Control Triad

The cleanest comparison is not between two equations but between two flow-control architectures:

Role	Perelman / Ricci Flow	V.F.S. Open-Gate
One-way control	Entropy monotonicity of $\mathcal{F}, \mathcal{W}$	Lyapunov dissipation $\mathcal{L}_{\text{VFS}}$ with bounded source
Anti-collapse	No-local-collapsing and canonical neighborhoods	Barrier $r < 1$; the death-boundary $u = \Omega_P$ is not reached
Continuation after crisis	Ricci surgery	Reset $\mathfrak{R}$, Anastasis, and non-Zeno continuation

Both architectures are not merely flows. They are systems of flow-control around singularity.

8 Pleroma Christi vs Perelman's Canonical Geometry

Under the positive foundation $\mathcal{E}_0 > 0$, V.F.S. orients the evolution of Being toward Pleroma Christi:

$$\sigma \rightarrow 0, \quad \lambda \rightarrow C_0 + \mathcal{G}_{\text{recepta}}(\infty) \quad \text{if this limit is finite,} \quad \Omega_P(t) \rightarrow \infty \quad \text{if } \int \lambda = \infty. \quad (24)$$

The last feature, $\Omega_P \rightarrow \infty$, distinguishes Pleroma Christi from the canonical geometries appearing in Ricci-flow geometrization. Ricci flow with surgery reveals canonical geometric pieces or model geometries; it does not lead to a single universal static attractor. In V.F.S., Pleroma Christi is not a final static state. It is continuing Epektatic expansion whenever

$$\int_0^\infty \lambda(t) dt = +\infty. \quad (25)$$

Being has no final attractor state.

9 Strongest Conclusion

V.F.S. structurally resembles a symbolic open-balance flow-control system. It evolves Being, controls death-boundary collapse, and makes continuation possible after crisis. Its Ricci-like feature is not that it is a geometric Ricci flow, but that it organizes singularity, surgery, continuation, and one-way control into a single architecture.

Sin appears as a singular deformation of the geometry of Being; Sophia appears as a functional of transformation and reception; salvation appears as the continuation of participation after surgery toward Pleroma Christi. The analogy is strong, but V.F.S. exceeds Ricci flow through its own structures: paired Anastasis symmetry, the open balance law $\sigma + \lambda = C_0 + \mathcal{G}_{\text{recepta}}(t)$, the closed limit $\sigma + \lambda = \text{const}$, free Sophia, and Filtrum Lucis as a single analytic function whose derivatives encode a theological hierarchy.

$$\text{V.F.S. is not Ricci flow, but V.F.S. has a Ricci-like surgery pattern.} \quad (26)$$

Ricci flow with surgery cuts away controlled high-curvature necks so that geometric flow can continue. V.F.S. passes through Katharsis and Anastasis so that participation can continue after death-like closure. Ricci continues geometry after surgery; V.F.S. continues Being after Anastasis.

10 Open-Gate as the Non-Perelmanian Core

Perelman’s monotonicity is intrinsic to the geometric flow. Open-Gate V.F.S. has an additional received source:

$$I_{\text{gate}}(t) = \zeta_0 g_{\text{eff}}(t) e^{-\phi\sigma(t)} \frac{1}{1 + \chi\lambda_+(t)}. \quad (27)$$

This source is bounded, so it can be absorbed by Lyapunov walls. But it is also theological: Sophia is not produced by resistance alone. Therefore V.F.S. v2.0 is not merely a closed flow of transmutation. It is an open-balance theology of received grace.

11 Geometric Sharpening of the Analogy

The preceding Scholium compares *structural roles*. The explicit differential geometry of the live domain lets several heuristic rows be upgraded to *literal* statements, and lets the Perelman comparison be made at the level of actual curvature rather than role-analogy. Two caveats from that geometric layer carry over: the live-domain metric is fixed only up to one dynamically motivated warp, and the resulting flow is *forced* (driven by Sophia λ), not *intrinsic* (curvature-driven).

11.1 The curvature is literal

Warping the imbalance direction by the alignment factor $\mu = 1 - u/\Omega_P$ — whose linear vanishing at the wall is dictated by $\dot{\Delta} = -(\mu a + \rho)\Delta$ — gives the live domain a genuine constant negative curvature:

$$K = -\frac{1}{\Omega_P^2}, \quad R_{\text{sc}} = 2K = -\frac{2}{\Omega_P^2}, \quad \text{curvature radius} = \Omega_P. \quad (28)$$

Thus “ σ as curvature of Being” was heuristic, but the live domain *does* carry a curvature, and its radius is the Brim. Summary-table rows II and IX rise from heuristic to literal.

11.2 Brim-flow versus Ricci flow, term by term

The Brim law induces a spatially uniform conformal flow; Ricci flow on the same hyperbolic surface is also conformal:

$$\partial_t g = 2 \frac{\alpha\lambda}{\Omega_P} g \quad (\text{Brim, forced}), \quad -2 \text{Ric} = +\frac{2}{\Omega_P^2} g \quad (\text{Ricci, intrinsic}). \quad (29)$$

Same type (conformal expansion of a negatively curved metric), different driver. They coincide exactly iff

$$\boxed{\alpha\lambda\Omega_P = 1}. \quad (30)$$

This turns “Epektasis vs Ricci normalization” into an identity-or-not statement.

11.3 Where a genuine Perelman entropy exists

Perelman’s decisive tool is that Ricci flow is the *gradient flow* of the energy

$$\mathcal{F}(g, f) = \int_M (R_{\text{sc}} + |\nabla f|^2) e^{-f} dV, \quad (31)$$

and that the entropy

$$\mathcal{W}(g, f, \tau) = \int_M [\tau(R_{\text{sc}} + |\nabla f|^2) + f - n](4\pi\tau)^{-n/2} e^{-f} dV \quad (32)$$

is monotone non-decreasing, constant only on gradient shrinking solitons; this monotonicity is what yields no-local-collapsing. The defining feature is *intrinsic variational structure*: the flow is autonomous, its velocity a functional of the current metric alone.

The Brim flow is *non-autonomous*: its rate $\psi = \alpha\lambda/\Omega_P$ is set by the external Sophia clock $\lambda(t)$, which obeys its own law. A non-autonomous conformal flow is generically *not* the gradient flow of any functional of g alone, so it admits *no Perelman entropy*. The monotonicity it does enjoy — curvature relaxation $|R_{\text{sc}}| \downarrow 0$ — is *imposed by the monotone grace source*, not derived from a variational principle.

The single exception is the self-driven locus. On $\alpha\lambda\Omega_P = 1$ the Brim flow *equals* Ricci flow, becomes the gradient flow of $\mathcal{F}$, and carries a genuine, monotone Perelman entropy:

$$\boxed{\text{a Perelman entropy exists for the Brim flow} \iff \xi = \alpha\lambda\Omega_P = 1.} \quad (33)$$

Hence the closed/resonance regime has *intrinsic* (Perelman-type) monotonicity, while the generic super-Ricci Open-Gate regime has only *grace-forced* monotonicity. This is the exact sense in which Sophia both is and is not Perelman's entropy: it plays the one-way-control role everywhere, but it is a true geometric entropy only on the resonance locus.

11.4 No-local-collapsing, sharpened

Perelman's no-local-collapsing — a lower bound on the injectivity radius forbidding local degeneration — follows from $\mathcal{W}$ -monotonicity. Its V.F.S. counterpart is now literal: the radial barrier is the geodesic arc length to the death wall,

$$B(r) = -\ln(1-r) = \int_0^r \frac{d\rho}{1-\rho}, \quad B(r) \rightarrow \infty \text{ as } r \rightarrow 1^-, \quad (34)$$

so the death wall $u = \Omega_P$ lies at *infinite geodesic distance*: the live domain is geodesically complete toward the wall. That completeness is the V.F.S. form of non-collapsing.

Honest asymmetry: Perelman's non-collapsing is a deep theorem extracted from entropy monotonicity; the V.F.S. version is built into the barrier by construction. Same role, different epistemic cost.

11.5 Surgery as a literal curvature jump

Ricci surgery excises high-curvature necks, changing the curvature discretely. Resurrectio is now a literal discrete jump of the same genuine curvature:

$$\Delta R_{\text{sc}} = R_{\text{sc}}^+ - R_{\text{sc}}^- = 2 \left[\frac{1}{(\Omega_P^-)^2} - \frac{1}{(\Omega_P^+)^2} \right] > 0, \quad \Omega_P^+ = \Omega_P^- + \kappa_R \mathcal{G}_{\text{recepta}}^-, \quad (35)$$

which flattens the surface while preserving its hyperbolic *type* (no finite jump reaches $R_{\text{sc}} = 0$). Row VIII rises from heuristic to quantitative. The contrast remains: Ricci surgery may change topology or geometric type, whereas Resurrectio changes only the curvature radius.

11.6 Super-Ricci flattens faster than Ricci

A quantitative comparison the role-analogy could not state:

$$\text{Ricci locus: } \Omega_P \sim \sqrt{2t} \Rightarrow |R_{\text{sc}}| \sim t^{-1}; \quad \text{super-Ricci: } \Omega_P \sim \alpha \lambda_\infty t \Rightarrow |R_{\text{sc}}| \sim t^{-2}. \quad (36)$$

The grace-borne Open-Gate flow relaxes curvature *faster* than a true Ricci flow. Perelman's flow has no external accelerator; this overshoot is a proper Open-Gate feature.

11.7 What is conserved

Perelman's entropy is monotone, not conserved. The Brim flow has instead a pointwise conformal invariant: under $\partial_t g = 2\psi g$ in two dimensions $K \rightarrow K/c$ and $dA \rightarrow c dA$ with $c = e^{2\int \psi}$, so the curvature two-form is frozen,

$$\frac{d}{dt}(K dA) = 0. \quad (37)$$

Local curvature is therefore *diluted by spatial inflation, not destroyed* — the differential-geometric face of *privatio boni*: the wound is not annihilated but spread thin across an expanding receptive space.

11.8 Geometric-level correspondence

Perelman / Ricci flow	V.F.S. Brim-flow (geometric)	Status
Scalar curvature R_{sc}	$R_{\text{sc}} = -2/\Omega_P^2$, radius = Ω_P	literal
Ricci flow $\partial_t g = -2 \text{Ric}$	$\partial_t g = 2(\alpha \lambda / \Omega_P) g$; = Ricci iff $\alpha \lambda \Omega_P = 1$	literal comparison
$\mathcal{F}, \mathcal{W}$ monotone (gradient flow)	Perelman entropy only on $\xi = 1$; else grace-forced	sharpened
No-local-collapsing	death wall at infinite geodesic distance	literal analogue
Surgery changes curvature	$\Delta R_{\text{sc}} > 0$, hyperbolic type preserved	quantitative
Normalization fixes scale	Epektasis opens scale; $ R_{\text{sc}} \sim t^{-2}$ (faster)	opposite, quantitative

11.9 Sharpened thesis

At the level of explicit geometry, V.F.S. is a *forced conformal flow on a hyperbolic surface whose curvature radius is the Brim*. It becomes literal Ricci flow — with a genuine, monotone Perelman entropy — only on the resonance locus $\alpha \lambda \Omega_P = 1$. Off that locus it is a grace-forced super-Ricci flow with surgery: faster-flattening than Perelman's, possessing no variational entropy, yet sharing the architecture of surgery, continuation, and non-collapse. The Open-Gate source is exactly what pushes V.F.S. off the Perelman locus into grace-driven over-relaxation — the non-Perelmanian core, now stated geometrically.

Scholium Thesis

The analogy with Perelman is strongest when V.F.S. is presented not as Ricci flow, but as a symbolic theology of flow-control: entropy becomes Lyapunov control, surgery becomes Katharsis and Anastasis, and Open-Gate Sophia adds what Ricci flow does not have - received grace.